

Equivariance Begins at the Source: Source Compatibility in Generative Robot Policies

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Abstract—Equivariant transport alone does not guarantee an equivariant manipulation policy: the source distribution must also transform consistently with the observation. We show that under an exact invertible equivariant flow, the policy inherits the source’s equivariance error and therefore cannot correct an incompatible source. In particular, the commonly used isotropic Gaussian centered at the world-frame origin is not translation-equivariant, with KL divergence growing quadratically with translation distance. We characterize compatible sources: no observation-independent probability distribution on $\mathbb{R}^3$ is translation-equivariant, while normalized Haar measure is the unique observation-independent equivariant distribution on $SO(3)$. A planar $SE(2)$ experiment validates the predicted equivariance errors for different sources.

I. INTRODUCTION

Diffusion and flow-matching models have become central to visuomotor imitation [1–3], motivating $SE(3)$ -equivariant denoisers [4, 5] and velocity fields [6]. However, equivariant transport alone does not guarantee an equivariant policy: the source distribution must also transform consistently with the observation [7, 8].

Recent work treats the source distribution as a design choice to improve generative policy performance [9–11], but its equivariance properties remain largely unexplored. In standard diffusion, the forward process typically fixes an isotropic Gaussian source, although observation-conditioned alternatives are possible [12, 13]. Flow matching instead makes the source a free design choice and generation an invertible transport. Under an exact equivariant flow, the policy therefore inherits the source’s equivariance error exactly.

We ask which source distributions are compatible with equivariant generative policies, rather than which source performs best. To isolate this question, we assume that the demonstration distribution is equivariant under joint transformations of observations and absolute actions. Real tasks may violate this assumption because of gravity, support surfaces, occlusions, or workspace constraints [14]; our analysis concerns the equivariant regime.

Our contributions are threefold. (1) We prove that equivariant generation cannot increase source equivariance error, and that exact invertible flows preserve it exactly. (2) For an observation-independent isotropic Gaussian, we derive a closed-form KL divergence that grows quadratically with translation distance and vanishes only for rotations about the source center. (3) We characterize all compatible sources through a canonical-frame representation, proving that no observation-independent distribution on $\mathbb{R}^3$ is translation-equivariant while normalized Haar measure is the unique

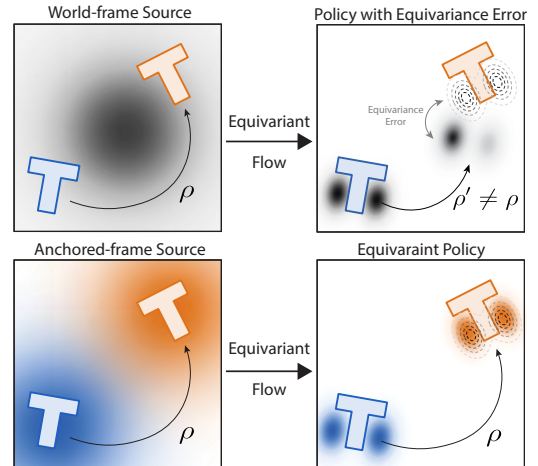


Fig. 1: A flow-matching policy predicts where to place the two fingertips to grasp a T-shaped object, one blob per fingertip. The model is trained with the T in the blue pose; the orange pose is unseen, obtained by transforming the blue one by ρ . An equivariant policy must transform its prediction by the same ρ (dashed). **Top:** with a fixed world-frame Gaussian source (gray), independent of the observation, the prediction for the orange T (grey) deviates from the dashed one, and the gap is the equivariance error inherited from the source. **Bottom:** with a source anchored at the object (trained with the blue distribution, evaluated with the orange one), the prediction coincides exactly with the dashed one.

observation-independent equivariant distribution on $SO(3)$, and derive practical constructions based on equivariant anchors and canonicalization [15–17].

II. RELATED WORK

Equivariant generative models and policies. For unconditional generative models, compatible sources combined with equivariant transport yield equivariant distributions [7]. Molecular models often remove translation by centering or quotienting coordinates [8, 12, 18], which is not directly available for absolute world-frame actions. Equivariant robot policies instead restrict guarantees to subgroups [4, 6] or handle translation through canonicalization, relative representations, or frame transfer [5, 19–21]. Other work studies symmetry breaking in data [22] or errors attributable to the learned field [23]. We instead isolate the source distribution and show that, under an exact invertible equivariant flow, its equivariance error is inherited exactly by the policy.

Source distributions as a design variable. Flow matching naturally supports arbitrary source distributions [24], and recent methods exploit this flexibility to improve generation or policy performance [9–11]. For example, WarmPrior initializes from action histories [10], while LeaP learns

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a proprioception-conditioned Gaussian [11]. These works optimize the source for path geometry or task performance; we instead study its equivariance properties and characterize compatible sources through an equivariant frame.

Canonicalization and frames. Symmetry can also be imposed through frame averaging [25], learned or slice-based canonicalization [16, 17, 26], stochastic symmetrization [27, 28], and probabilistic symmetry breaking [15]. These approaches enforce symmetry through coordinate transformations or distribution-level constructions, rather than solely through equivariant dynamics. Scene-derived frames may face continuity obstructions [29], while proprioceptive frames can be obtained directly, though their usefulness depends on the task [30]. Our canonical-frame characterization shows that, given an equivariant frame, every compatible source can be represented in the canonical frame and transformed back to the world frame.

III. PRELIMINARIES

A. Problem setup.

We frame visuomotor manipulation as imitation learning from expert demonstrations. The observation $O = (X, s)$ combines a partial scene point cloud $X \in \mathbb{R}^{N \times 3}$ with proprioception $s = (T^{\text{ee}}, g)$, where $T^{\text{ee}} = (R^{\text{ee}}, p^{\text{ee}}) \in SE(3)$ is the end-effector pose and $g \in \{0, 1\}$ the gripper state. The action $a = (T^a, g^a) \in SE(3) \times \{0, 1\}$ specifies an *absolute world-frame* target pose and gripper command.

B. Group action and the equivariance target.

For $\rho = (R, v) \in SE(3)$, rigid transformations act componentwise as

$$\rho \cdot x = Rx + v, \quad x \in X, \quad \rho \cdot T^{\text{ee}} = (RR^{\text{ee}}, Rp^{\text{ee}} + v), \quad \rho \cdot g = g,$$

and analogously on actions. Writing $\pi = (\pi_T, \pi_g)$, we require $\pi_T(\cdot \mid \rho \cdot O) = \rho_{\#} \pi_T(\cdot \mid O)$ and $\pi_g(\cdot \mid \rho \cdot O) = \pi_g(\cdot \mid O)$ for all $\rho \in SE(3)$.

C. Conditional flow matching.

Conditional flow matching (CFM) generates actions by solving $\frac{d}{dt} z(t) = v_{\theta}(z(t), t, O)$ with $z(0) \sim p_0(\cdot \mid O)$, where the learned velocity field transports the source p_0 toward the demonstrated action distribution [2, 31, 32]. For the straight path $z_t = (1-t)z_0 + tz_1$, CFM regresses $v_{\theta}(z_t, t, O)$ toward $z_1 - z_0$. The resulting policy is the pushforward of p_0 through the learned flow. Unlike standard diffusion, flow matching leaves the source distribution as a design choice [9, 31, 32].

D. Conditions for Policy Equivariance

An equivariant generative policy requires both a compatible source and an equivariant generation mechanism [5, 7, 8]. Let K_O denote the generation kernel from a source sample z to an action, so that $p_{\theta}(\cdot \mid O) = \int K_O(\cdot \mid z) p_0(dz \mid O)$. Sufficient conditions for policy equivariance are

$$(A1) \quad p_0(\rho \cdot A \mid \rho \cdot O) = p_0(A \mid O),$$

$$(A2) \quad K_{\rho \cdot O}(\rho \cdot B \mid \rho \cdot z) = K_O(B \mid z),$$

for measurable sets A and B ; equivalently, $p_0(\cdot \mid \rho \cdot O) = \rho_{\#} p_0(\cdot \mid O)$ and $K_{\rho \cdot O}(\cdot \mid \rho \cdot z) = \rho_{\#} K_O(\cdot \mid z)$. Here, A1 requires source compatibility, while A2 requires equivariance of the generation mechanism. For flow matching, A2 holds when, $v_{\theta}(\rho \cdot z, t, \rho \cdot O) = \rho_{\#} v_{\theta}(z, t, O)$, where $\rho_{\#}$ is the differential of $z \mapsto \rho \cdot z$.

Most equivariant robot policies focus on A2 using equivariant backbones such as $SE(3)$ -Transformers or GATr [5, 33–37], while retaining an isotropic Gaussian source [4, 38]. Section IV therefore focuses on A1 and asks: (Q1) what equivariance error does the default Gaussian induce? (Q2) which common sources satisfy A1?

IV. QUANTIFYING AND CHARACTERIZING SOURCE DISTRIBUTIONS

A. Propagation of Equivariance Error

We now quantify the effect of violating A1 when A2 holds. For a conditional distribution $q(\cdot \mid O)$, define its *equivariance error* at (ρ, O) as

$$\mathcal{E}_{\rho, O}[q] = \text{TV}(q(\cdot \mid \rho \cdot O), \rho_{\#} q(\cdot \mid O)),$$

so that $\mathcal{E}_{\rho, O}[q] = 0$ exactly when q is equivariant at (ρ, O) .

Theorem 1 (Equivariance-error propagation). *Assume A2. Then for every $\rho \in SE(3)$ and observation O : (i) $\mathcal{E}_{\rho, O}[p_{\theta}] \leq \mathcal{E}_{\rho, O}[p_0]$; (ii) if K_O is induced by an invertible map, as for an exact ODE flow, then $\mathcal{E}_{\rho, O}[p_{\theta}] = \mathcal{E}_{\rho, O}[p_0]$.*

Proof. By A2, the two measures compared in $\mathcal{E}_{\rho, O}[p_{\theta}]$ are obtained by applying the same Markov kernel $K_{\rho \cdot O}$ to the two measures compared in $\mathcal{E}_{\rho, O}[p_0]$. Data processing therefore gives (i), while an invertible pushforward preserves TV and gives (ii). The same argument applies to any f -divergence, including KL. $\square$

Thus, an equivariant generator cannot increase the source’s equivariance error, and an exact invertible flow preserves it exactly. In particular, a compatible source paired with equivariant transport yields an equivariant policy, while under an exact equivariant flow any source incompatibility is inherited unchanged by the policy. With a discretized ODE solver, the contraction result still holds when A2 is preserved, but exact equality need not.

B. Equivariance Error of the Gaussian Source

We now apply Theorem 1 to the commonly used observation-independent source, $p_0 = \mathcal{N}(0, \sigma^2 I)$, in world coordinates. Since the theorem also holds for KL divergence, its equivariance error can be computed in closed form.

Corollary 1 (Gaussian source equivariance error). *For every $\rho = (R, v) \in SE(3)$, $\text{KL}(p_0 \parallel \rho_{\#} p_0) = \|v\|^2 / 2\sigma^2$. Under an exact equivariant flow, the policy has the same KL equivariance error. The error depends only on translation and vanishes iff $v = 0$; thus, the fixed Gaussian is equivariant only to rotations about its center.*

Increasing σ reduces the error as $1/\sigma^2$, but no finite variance eliminates it. This motivates the next question: which source distributions satisfy A1 exactly?

C. Characterizing Compatible Sources

We now characterize all source distributions that satisfy A1. An *equivariant frame* is a Borel map τ from observations to $SE(3)$ satisfying $\tau(\rho \cdot O) = \rho \tau(O)$. In our setting, the end-effector pose provides one such frame. The corresponding canonicalized observation is $M_\tau(O) = \tau(O)^{-1} \cdot O$ which expresses O relative to this frame; the proposition below shows that every compatible source can be specified in these canonical coordinates and transformed back through $\tau(O)$.

Proposition 1 (Canonical-frame characterization). *Let $\mathcal{C}_\tau = \{O : \tau(O) = e\}$ denote the canonical slice. A source distribution p_0 satisfies A1 if and only if there exists a Markov kernel $\tilde{q}$ on $\mathcal{C}_\tau$ with*

$$p_0(\cdot | O) = \tau(O)_\# \tilde{q}(\cdot | M_\tau(O)) \quad \text{for all } O.$$

For a fixed frame τ , the kernel $\tilde{q}$ is unique.

Proof. Suppose first that the stated representation holds. Since frame equivariance gives $\tau(\rho \cdot O) = \rho \tau(O)$ and $M_\tau(\rho \cdot O) = M_\tau(O)$,

$$p_0(\cdot | \rho \cdot O) = (\rho \tau(O))_\# \tilde{q}(\cdot | M_\tau(O)) = \rho_\# p_0(\cdot | O),$$

which is equivalent to A1.

Conversely, suppose p_0 satisfies A1 and define $q(\cdot | O) := (\tau(O)^{-1})_\# p_0(\cdot | O)$. Using A1 and frame equivariance gives $q(\cdot | \rho \cdot O) = q(\cdot | O)$. Taking $\rho = \tau(O)^{-1}$ yields $q(\cdot | O) = q(\cdot | M_\tau(O))$. Thus, defining $\tilde{q} := q|_{\mathcal{C}_\tau}$ gives $p_0(\cdot | O) = \tau(O)_\# \tilde{q}(\cdot | M_\tau(O))$. Finally, for $c \in \mathcal{C}_\tau$, $\tau(c) = e$ and $M_\tau(c) = c$ so $\tilde{q}(\cdot | c) = p_0(\cdot | c)$, which proves uniqueness. $\square$

The constructive direction is classical [39, 40]; our converse shows that, given an equivariant frame, every compatible source has this canonical-frame representation. Thus, designing a compatible source amounts to choosing the canonical distribution $\tilde{q}$ and what observation information it may depend on.

To make the resulting design space concrete, decompose the action as $a = (t, Q, g)$ into translation, rotation, and gripper components, with $\rho \cdot (t, Q, g) = (Rt + v, RQ, g)$. If the source factorizes across these components given O , then it satisfies A1 if and only if each factor does. Without this factorization, equivariant marginals alone do not guarantee that the joint source satisfies A1. This reduction leads to a sharp distinction between translation and rotation.

Proposition 2 (Translation-rotation dichotomy). *(i) No observation-independent probability measure on $\mathbb{R}^3$ is translation-equivariant. (ii) On $SO(3)$, the normalized Haar measure is the unique observation-independent rotation-equivariant distribution.*

Proof sketch. (i) A translation-invariant probability measure would assign equal mass to each translated unit cube in a countable tiling of $\mathbb{R}^3$, contradicting countable additivity. (ii) Uniqueness of the normalized Haar measure follows from Haar’s theorem for the compact group $SO(3)$. $\square$

TABLE I: Representative compatible-source constructions. Columns are implementation mechanisms and may realize the same kernel; they are neither disjoint nor exhaustive parameterizations of $\tilde{q}$.

	(a) Observation-Independent	(b) Anchored	(c) Canonicalization-Induced
Rotation	Haar (unique)	$m(O) \cdot \kappa_R$	Transform back through $\tau(O)$
Translation	None exists	$b(O) + \kappa_T$	

Claim (ii) concerns distributions directly on $SO(3)$; in practice, Haar rotations can be generated by applying Gram–Schmidt to isotropic Gaussian samples in the 6D rotation representation. By contrast, a uniform distribution over a fixed workspace is tied to world coordinates and is therefore not globally translation-equivariant. Thus, translation necessarily requires observation dependence, whereas rotation does not. We next ask how much observation dependence is sufficient when it is required.

Corollary 2 (Equivariantly anchored sources). *Let $b(O) \in \mathbb{R}^3$ be an equivariant point, $b(\rho \cdot O) = Rb(O) + v$, and let $m(O) \in SO(3)$ be an equivariant orientation, $m(\rho \cdot O) = Rm(O)$. (i) If $\epsilon \sim \kappa_T$, where κ_T is rotation-invariant on $\mathbb{R}^3$, then $b(O) + \epsilon$ defines a compatible translation source. (ii) If $S \sim \kappa_R$ for any distribution κ_R on $SO(3)$, then $m(O)S$ defines a compatible rotation source.*

The asymmetry follows from the group action. For translation, the anchor $b(O)$ tracks the translation v but the noise is still rotated by R , so κ_T must be rotation-invariant. For rotation, the orientation $m(O)$ absorbs the entire left action of R leaving κ_R unconstrained. Table I summarizes these constructions.

Proposition 1 also recovers canonicalization: observations and sources are expressed in a canonical frame and transformed back through $\tau(O)$ [5, 15–17]. In Table I, this corresponds to column (c), while equivariant anchored sources correspond to column (b).

V. EXPERIMENT

Section IV gives exact predictions for source-induced equivariance error. We test these predictions in a planar $SE(2)$ setting, where all relevant quantities can be computed directly. All model components are held fixed across experiments except the source distribution, isolating its effect on the policy’s equivariance error.

World and demonstrations. The action is a planar grasp location $a \in \mathbb{R}^2$. The observation $O = (o, q)$ consists of an object position o and a unit heading vector q , $\|q\| = 1$, with $\rho = (R, v) \in SE(2)$ acting by $(o, q) \mapsto (Ro + v, Rq)$. This defines the equivariant frame $\tau(O) = (o, R_o)$, where $R_o = [q, Jq]$ and $J = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ rotates vectors by 90° . The demonstration distribution is an asymmetric bimodal Gaussian mixture in the object frame, with weights (0.7, 0.3), means $(1.0, \pm 0.6)$, and covariance $0.15^2 I$, transported to world coordinates by $\tau(O)$. The asymmetry avoids accidental

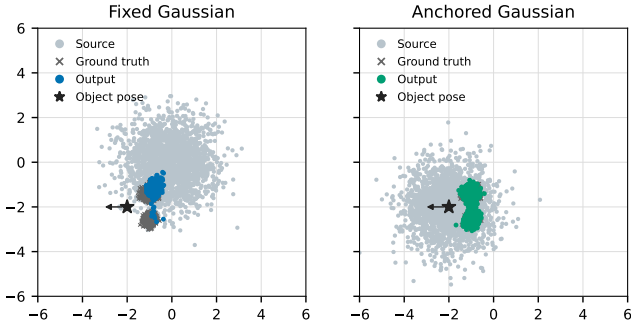


Fig. 2: Prediction on an unseen scene. Predictions from models trained with the fixed and the anchored Gaussian sources for an unseen observation $O = (o, q)$ with $o = (-2, -2)$ and $q = (-1, 0)$. The anchored-source prediction more closely follows the demonstrated target distribution in this example.

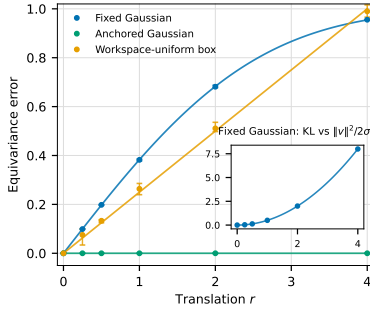


Fig. 3: Equivariance error under translation. TV equivariance error as the scene is translated by r . Markers show errors measured from the learned policies and curves show the corresponding closed-form source-side predictions; the inset compares the fixed Gaussian’s KL divergence with the quadratic $\|v\|^2/2\sigma^2$ law. The fixed Gaussian and workspace-uniform source incur increasing error with translation, while the anchored Gaussian remains equivariant.

symmetries, while bimodality makes the learned transport nontrivial. Training uses $\sim 10^5$ scenes with o sampled uniformly from $[-2, 2]^2$ and the heading uniformly from $[0, 2\pi)$.

One field, three sources. All source conditions use the same frame-conjugated velocity field architecture $v_\theta(z, t, O) = R_o f_\theta(R_o^\top(z - o), t)$, where f_θ is a small MLP trained with conditional flow matching using the same architecture, optimizer, and training budget. This parameterization satisfies A2 for *every* θ ; therefore, by Theorem 1(ii) the policy’s equivariance error is determined by the source. The conditions differ only in $p_0(\cdot | O)$, with three training seeds each:

- **Fixed Gaussian** $\mathcal{N}(0, I)$: the observation-independent standard practice analyzed by Corollary 1;
- **Anchored Gaussian** $\mathcal{N}(o, I)$: the compatible source of Corollary 2 anchored at the object position;
- **Workspace-uniform box** $U(B)$, $B = [-2, 2]^2$: an observation-independent bounded source fixed to the workspace.

Results. Figure 2 shows predictions on a scene unseen during training. The anchored Gaussian better captures the demonstrated target distribution than the fixed Gaussian,

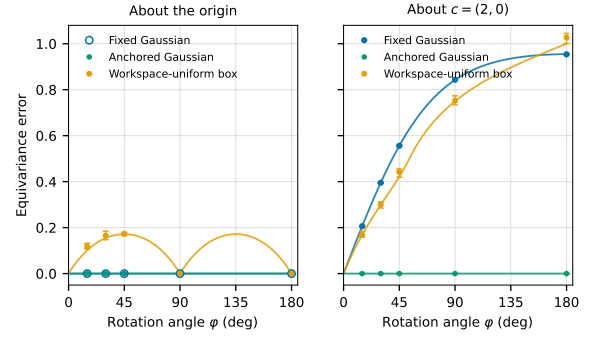


Fig. 4: Equivariance error under rotation. TV equivariance error as the scene is rotated by φ about the origin (left) and about the off-center point $c = (2, 0)$ (right); markers and curves as in Fig. 3. The fixed Gaussian is equivariant for rotations about its center but not about c ; the workspace-uniform source is equivariant only at rotations that preserve the square workspace; the anchored Gaussian remains equivariant throughout.

qualitatively illustrating the effect of source compatibility.

Figure 3 measures equivariance error under translations $v = (r, 0)$, $r \in \{0, 0.25, 0.5, 1, 2, 4\}$, over eight unseen scenes. The measured policy errors closely follow the closed-form source-side predictions (for the Gaussian, $TV = 2\Phi(\|v\|/2\sigma) - 1$ with Φ the standard normal cdf). The fixed Gaussian and workspace-uniform source incur increasing error with translation, while the anchored Gaussian remains equivariant throughout. In particular, the fixed Gaussian follows the quadratic KL law of Corollary 1, whereas the bounded workspace-uniform source remains incompatible despite covering the training workspace.

Figure 4 shows the corresponding results under rotations by $\varphi \in \{15^\circ, 30^\circ, 45^\circ, 90^\circ, 180^\circ\}$ about the origin and the off-center point $c = (2, 0)$. The fixed Gaussian has zero error for rotations about its own center, but becomes non-equivariant when the rotation center is displaced. The workspace-uniform source has zero error only at rotations that preserve the square workspace, while the anchored Gaussian remains equivariant in both settings.

Across the translation and rotation experiments, the measured policy errors follow the source-side predictions, consistent with Theorem 1(ii): an exact equivariant flow preserves the source’s equivariance error.

VI. CONCLUSION

An exact invertible equivariant flow preserves the equivariance error of its source distribution, so source compatibility is necessary for an equivariant flow-matching policy. We quantify this error for the standard world-frame Gaussian and characterize all compatible sources through a canonical-frame representation, yielding practical constructions based on equivariant anchors and canonicalization. Experiments on a planar $SE(2)$ system show that the measured policy errors follow the predicted source-side errors. Together, these results identify the source distribution as an explicit design consideration for equivariant generative policies.

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